

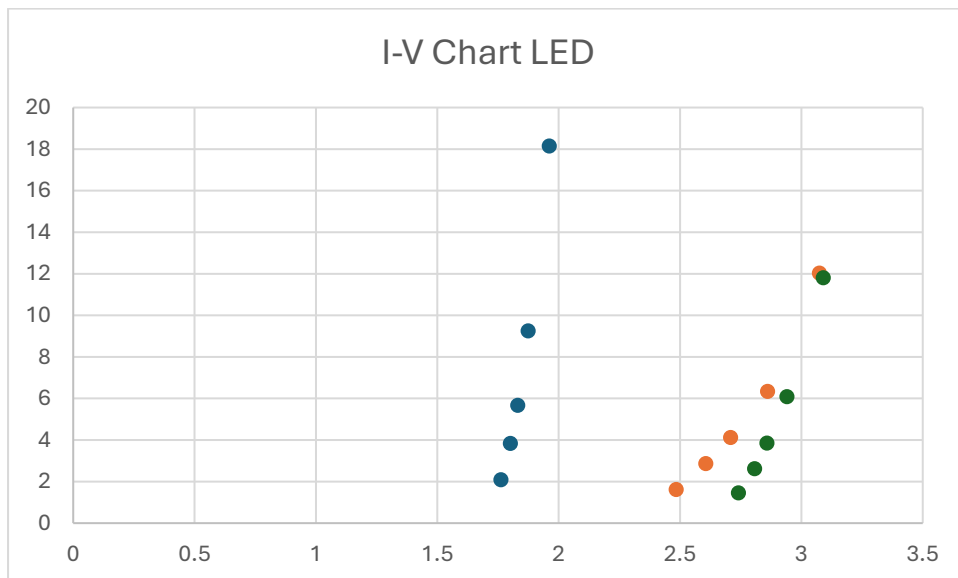
EEP2 – Week 1

17/1/2025 – E4A-06-03

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Activity 1a

		Values for R_R , R_G and R_B				
Nominal Resistance →		150 Ω	330 Ω	560 Ω	820 Ω	1500 Ω
Measured Resistance →		149.2	331.2	547	819	1489
Red LED	Voltage V_R	1.962	1.875	1.832	1.802	1.763
	Current I_R	18.14	9.24	5.66	3.83	2.08
Green LED	Voltage V_G	3.075	2.861	2.709	2.607	2.485
	Current I_G	12.02	6.33	4.11	2.86	1.61
Blue LED	Voltage V_B	3.091	2.941	2.859	2.808	2.741
	Current I_B	11.8	6.08	3.84	2.61	1.45

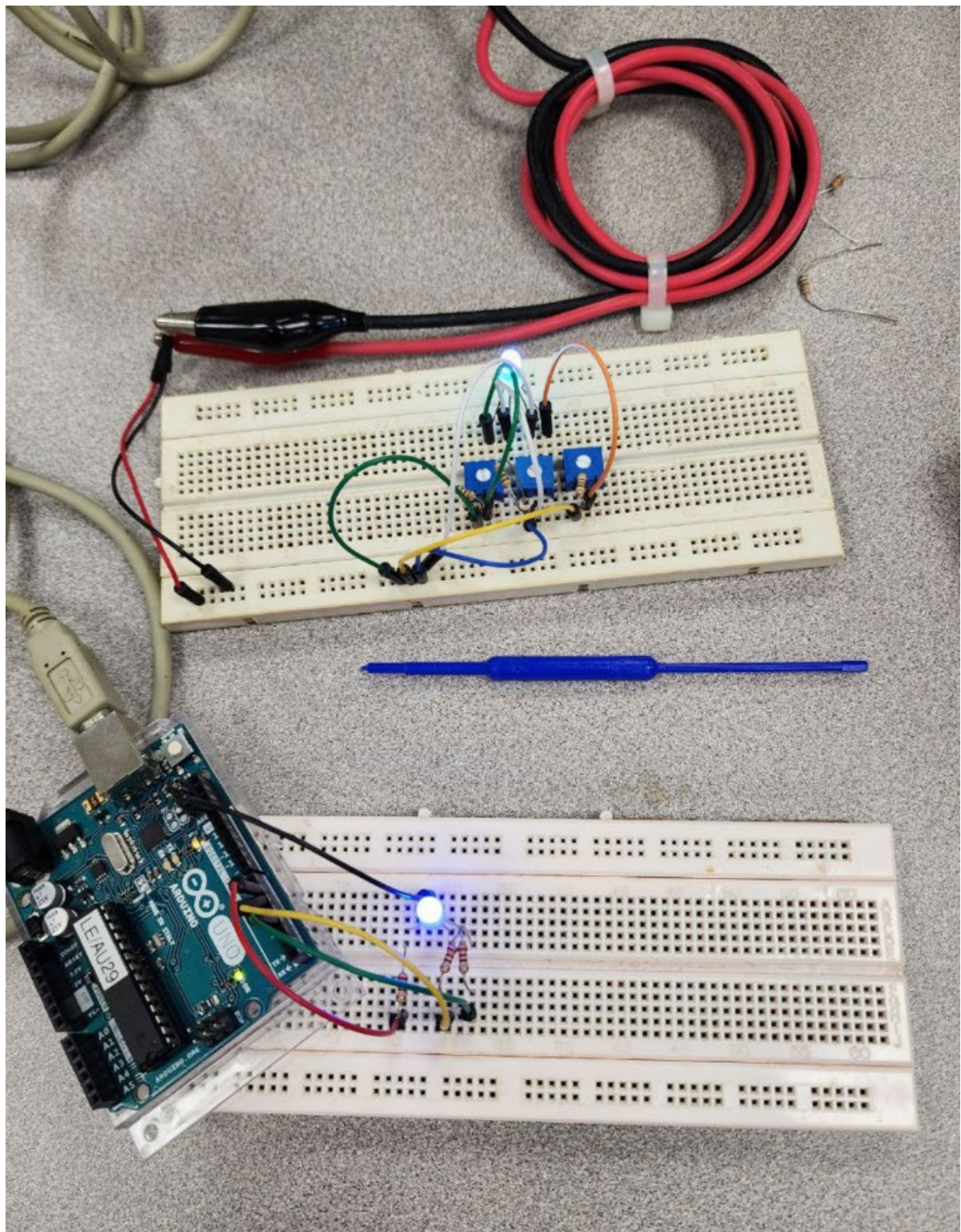


Activity 1b

B - I_R : 20 mA, I_G - 1.5 mA, I_B - 0 mA

		Values for R_R , R_G and R_B				
Nominal Resistance →		150 Ω	330 Ω	560 Ω	820 Ω	1500 Ω
Measured Resistance →		149.2	331.2	547	819	1489
Red LED	Voltage V_R	1.962	1.875	1.832	1.802	1.763
	Current I_R	18.14	9.24	5.66	3.83	2.08
Green LED	Voltage V_G	3.075	2.861	2.709	2.607	2.485
	Current I_G	12.02	6.33	4.11	2.86	1.61
Blue LED	Voltage V_B	3.091	2.941	2.859	2.808	2.741
	Current I_B	11.8	6.08	3.84	2.61	1.45

Resulted color: Yellow



Arduino

Make wave like smooth transition

```
const int redPin = 6;
```

```
const int bluePin = 5;
```

```
const int greenPin = 3;
```

```
const int delayTime = 10;
```

```
const int stepSize = 5;
```

```
void setup() {
```

```
    pinMode(redPin, OUTPUT);
```

```
    pinMode(greenPin, OUTPUT);
```

```
    pinMode(bluePin, OUTPUT);
```

```
}
```

```
void loop() {
```

```
    smoothTransition(255, 0, 0, 0, 255, 0);
```

```
    smoothTransition(0, 255, 0, 0, 0, 255);
```

```
    smoothTransition(0, 0, 255, 255, 0, 0);
```

```
    smoothTransition(255, 0, 0, 255, 150, 0);
```

```
    smoothTransition(255, 150, 0, 255, 50, 100);
```

```
}
```

```
void smoothTransition(int startR, int startG, int startB, int endR, int endG, int endB) {
```

```
    int steps = 255 / stepSize;
```

```
    for (int i = 0; i <= steps; i++) {
```

```
        int redValue = map(i, 0, steps, startR, endR);
```

```
        int greenValue = map(i, 0, steps, startG, endG);
```

```
        int blueValue = map(i, 0, steps, startB, endB);
```

```
        analogWrite(redPin, redValue);
```

```
analogWrite(greenPin, greenValue);
```

```
analogWrite(bluePin, blueValue);
```

```
delay(delayTime);
```

```
}
```

```
}
```